Performing a *t*-Test

One way to assess whether the results of an experiment are statistically significant is to perform a *t*-test. Consider an experiment in which one group of bean plants was treated with fertilizer, whereas a control group was not. Before beginning the experiment, the researchers hypothesized that fertilizer would not affect plant height (the fertilizer would neither increase plant height nor decrease it).

When the experiment was completed, the fertilized plants appeared overall to have grown taller than the unfertilized ones—that is, the mean height of the fertilized plants was greater than the mean height of the unfertilized ones. This result suggests that the fertilizer did, in fact, have an effect and, hence, that the two means are not equal. However, it is also possible that the different means in the two study groups resulted from natural variation of plant heights within the two groups, particularly if the total number of plants is small. How can we determine the likelihood that the observed differences are meaningful and therefore indicate that the fertilizer had an effect? The *t*-test provides a standardized way to decide whether the fertilizer had a significant effect on mean plant height.

To perform a *t*-test, the first step is to calculate the value T (named for "*t*-test"):

$$T = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{(s_1^2 + s_2^2)}{n}}}$$

In this equation, $\bar{x}_1$ is the mean for the experimental group (fertilized plants), $\bar{x}_2$ is the mean for the control group (unfertilized plants), s_1 is the standard deviation for the experimental group, and s_2 is the standard deviation for the control group. Finally, n is the number of observations in each of the groups. [Note: The formula shown here is valid when the experimental and control groups have the same number of observations (n). A different formula would be used if the two groups had different numbers of observations.]

To calculate T, plug in the values for $\bar{x}_1$, $\bar{x}_2$, s_1, s_2, and n. T will be close to zero when the means $\bar{x}_1$ and $\bar{x}_2$ are nearly equal, and T will be farther from zero (will differ more greatly from zero) when the means are considerably different.

Is the calculated value T different enough from zero to reject the hypothesis that the two means are equal? This decision is based on the probability (p) that an observed difference between two means could have occurred simply by chance (assuming the initial hypothesis was correct, that is, that the two means are equal). The value of p can be determined using a t distribution that has $2(n-1)$ degrees of freedom, where n is the number of observations. When p is small (typically, when it is less than 0.05), we reject the hypothesis that the means $\bar{x}_1$ and $\bar{x}_2$ are equal. The value of p can be obtained from an online calculator or looked up in tables for t distributions in a statistics textbook.